

Interface Capability Brief

eiConsole for Healthcare – HL7 Demo · Stakeholder overview · 2026-08-31

1. Executive summary

Show the eiConsole for Healthcare route grid exactly as the website does: LLP Listener, HL7 2.x to XML, logical map of last name / first name / date of birth, one Target, SQL mapper, database Transport.

2. Who this is for

- Sources: Hospital ADT over HL7 LLP (TCP) - Destinations: data-measures SQL Server (dbo.Patients) - Demo vs production: synthetic ADT only. No SNIP, no real MPI.

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3. What the interface can do

Capabilities below are drawn from the working design and, when present, the published CapabilityStatement — not from aspirational roadmap items.

- Transport: HL7 LLP (HL7TCPLListener) host port 2578
- Format: HL7 v2.5.1 ADT^A01 (pipe-delimited), MLLP framed
- Identity: MSH-10 control ID
- Samples: samples/hospital_ADT_A01.hl7

Integration highlights

FQCN sources: public demo + PilotFish_V2 modules.conf + Med Rec HL7 to XML format + csv-sftp-to-sql / lab LLP skeletons. Image: pilotfish-eip:23R1.

Stage	Module / mechanism	Notes
Listener	com.pilotfish.eip.modules.hl7.HL7TCPLListener	Port 2578, Engine=PilotFish, async AA
Format profile	Relay (System Format)	23R1 runtime; website stages are the processors below
Source Transform	HL7TransformationProcessor	HL7 2.X to XML (website dropdown)
Logical map	XSLTProcessor hl7-xml-to-patient.xslt	LastName, FirstName, DateOfBirth; PID.7 yyyyMMdd → yyyy-MM-dd
Routing	NullRoutingModule	Single Target (website: one database)
Target Transform	XSLTProcessor patient-to-sqlxml.xslt	Patient XML → SQLXML Insert
Transport	com.pilotfish.eip.modules.db.DatabaseSqlTransport	JDBC via \$\$\$sqlserver.*

4. Honest boundaries (not claimed yet)

- None for this website slice.

5. What you can verify in a walkthrough

Use these as checklist items when reviewing the demo with business stakeholders. They describe outcomes, not module names.

- Exercise route “1 - Hospital HL7 To Database” from the Web UI or documented smoke path
- Confirm outputs land in the documented folders / queues and appear in the Web UI status views

6. How it works (plain language)

7. Validation

- What is checked: LLP ACK, patient XML date format, SQL row - What is NOT checked: full ADT IG, MPI match - Failure blocks outbound: HL7 parse / XSLT throw

6. State & idempotency

- SQL insert is append-only. Resend duplicates (accepted). - Snapshots under output/snapshots/ match the website Testing Mode panes.

8. Dual-write / side effects

- Order: LLP → HL7 XML snapshot → patient XML → SQLXML snapshot → JDBC insert

3. Inbound contract

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Routes in this interface

Each route is a configured PilotFish flow. Technical wiring appears in the Route Diagrams PDF; here is the stakeholder reading.

1 - Hospital HL7 To Database: runs steps such as HL7 over LLP, HL7 2.x to XML, Snapshot HL7 XML, Map PID to Patient XML.

7. Security & trust posture

- X12 SNIP validation is used where configured to reject bad EDI before downstream work
- Credentials and passwords in this package are demo-only — not production secrets
- Risk noted in design: Med — HL7 XML shape varies with friendly names — Mapper XSLT must be path-tolerant
- Risk noted in design: Med — 23R1 vs 26R1 docs — Website may show newer UI
- Risk noted in design: Low — Website test starts after Listener — We send real MLLP so the Listener is in the path

8. Seeing it work

- - Ports: Web UI 8142, EIP 8141, LLP 2578, SQL 14342 - Credentials: sa / PilotFish_Demo1! (demo only) - Heap: 2GB - Cold start: SQL healthy + init, then EIP ~60–90s
- - Local Web UI: `http://127.0.0.1:8142/`
- - EIP: `http://127.0.0.1:8141/eip/`
- - SQL: `127.0.0.1:14342 sa / `PilotFish_Demo1!`
- `docker compose --profile full up -d --build`

9. Related technical documents

- DESIGN.md — working engineering specification for this interface
- README.md — how to run, ports, and smoke commands
- HL7_Interface_Engine_V2_Route_Diagrams.pdf — detailed PilotFish route wiring diagrams

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- This Capability Brief — shareable stakeholder summary (regenerate after design changes)

This document is generated from the interface sources listed above. If design and runtime diverge, believe the smoke-tested runtime and update DESIGN.md.